The Projection Law: Model-Ensemble Errors Point at Their Binding Constraint

Alex Welcing
Lupine Science
alex@lupine.io

Draft of June 11, 2026

Abstract

When many independently constructed models agree, the agreement is routinely read as confidence. We formalize and test the opposite reading: a model family is a projection operator, fitting drives every member toward the point of the family's reachable set nearest the truth, and the shared residual — one direction in observable space — is a fingerprint of whatever constraint binds the family. We prove the core as machine-checked theorems (Lean 4, zero unproven obligations): on any convex family, best approximations are unique and share one residual lying in the family's normal cone; the resulting bias-plus-noise ensembles have participation ratio $\mathrm{PR}(d,\rho) = (\rho+d)^2/((\rho+1)^2+(d-1))$, collapsing to 1 at explicit rate $3(d-1)/\rho$; and the participation ratio is provably blind to sign structure that pairwise alignment detects. We then test the law's sharpest consequence — *errors organize by constraint, not by implementation* — with pre-registered factorial experiments at three layers of one epistemic stack. Classical interatomic potentials (559 models): within-family error correlation 0.95, error dimensionality invariant over forty years. Foundation machine-learned potentials (8 MatPES models × 2 functionals × 4 architectures): errors cluster by training functional, not architecture (S_{func}=+0.317 vs. S_{arch}=−0.093, permutation p=0.029). DFT implementations (12 ACWF methods, 384 systems): errors cluster by pseudopotential table, not simulation code (S_{table}=+0.526 vs. S_{code}=+0.265, p=0.017). Each registered test carried an explicit refutation condition; neither was triggered. Across paradigm replacements the error anisotropy is conserved while its direction rotates to the next constraint upstream — a conservation law for the geometry of ignorance, with a practical corollary: a single shared correction vector removes ∼96% of foundation-model elastic-constant error, and cross-model alignment is a zero-cost detector of when such calibration is trustworthy.

1 Introduction

Every field that builds models in families uses inter-model agreement as evidence. The practice has long been questioned qualitatively: climate ensembles share construction and history, so their consensus may reflect common structure rather than truth (Pirtle et al., 2010; Parker, 2011), and error correlations between members quantify that dependence (Bishop and Abramowitz, 2013; Annan and Hargreaves, 2011; Knutti et al., 2013). In machine learning, independently trained networks agree far beyond what their accuracy predicts (Mania et al., 2019; Geirhos et al., 2020), ensemble members cluster in prediction space (Fort et al., 2019), and disagreement tracks the shared bias of the training procedure rather than correctness (Jiang et al., 2022). What these literatures establish qualitatively, this paper makes geometric, formal, and—in the specific sense of identifying *which* constraint binds—operational.

The central object is the *error vector*: for model i in family F evaluated on observables with reference values, $e_i \in \mathbb{R}^d$ is the vector of (relative) errors. The law has three parts:

(i) Projection. A model family is a projection operator. Fitting, however performed and by whomever, drives predictions toward the point of the family's reachable set nearest the truth; the residual is a property of the *(family, target)* pair, not of any individual model or modeler. All well-fitted members therefore share one residual direction (Theorems 1–2).

(ii) Gauge. The ensemble's error second moment is a shared bias plus fitting noise; its participation ratio is a closed-form gauge of the systematic fraction (Theorem 3), so low error dimensionality is not a curiosity but a necessity for any converged family, at an explicit rate.

(iii) Conservation and rotation. When a modeling paradigm is replaced — when the binding constraint is removed — the error anisotropy does not dissolve into isotropic scatter. It is conserved, and its direction rotates to the next constraint upstream in the epistemic stack. This is the part of the law for which we find no antecedent in the climate, forecasting, machine learning, or materials literatures, and it is the part we test hardest.

We instantiate the law at three layers of a single epistemic stack — classical interatomic potentials, foundation machine-learned interatomic potentials (MLIPs), and the density-functional-theory (DFT) implementations that generate their training data — because this stack offers what few domains can: large open model ensembles, factorial structure (the same architecture trained on two functionals; the same simulation code run with two pseudopotential families), and references at every layer. The empirical discovery paper for the first layer is reported separately (Welcing, 2026); here that layer serves as the observational base of a cross-layer test.

Three methodological commitments distinguish this work. The theory core is *machine-checked*: every theorem below is verified in Lean 4 with zero unproven obligations, and a written contract maps each theorem to the statistic that instantiates it. The two new experiments are *pre-registered*, with thresholds and explicit refutation conditions committed to version control before any model was executed; we report all threshold misses as failures. And the entire evidence chain is packaged for *tiered replication*: the analysis layer verifies from committed raw data with no machine-learning dependencies in seconds, and the physics layer re-derives the raw data from public checkpoints bit-exactly.

2 The formal core

The mathematics of best approximation is classical (Deutsch, 2001; Kinderlehrer and Stampacchia, 1980); we claim only its epistemic application to model ensembles and the machine-checked instantiation chain. Throughout, E is a real inner-product space (observable space), $s \subseteq E$ a model family's reachable set, $T \in E$ the truth.

Theorem 1 (Normal-cone criterion)**.** *Let s be convex and p a best approximation of T in s (i.e. $p \in s$ and $\|T - p\| \leq \|T - q\|$ for all $q \in s$). Then the residual $T - p$ lies in the normal cone of s at p: $\langle T - p, q - p\rangle \leq 0$ for every $q \in s$.*

Theorem 2 (Consensus)**.** *Best approximations onto a convex family are unique; consequently any two best approximations of the same target share an* identical *residual. The residual — hence any agreement among independently fitted members — is determined by the (family, target) pair alone, and vanishes exactly when the truth lies in the family.*

Agreement among models sharing a constraint is therefore a measurement of the constraint, not evidence of truth: the formal content of the qualitative warnings of Parker (2011) and Pirtle et al.

(2010), and a strengthening of error-correlation dependence measures (Bishop and Abramowitz, 2013) in that, combined with the factorial designs of §4–5, it identifies *which* constraint binds.

Theorem 3 (Gauge)**.** *Model errors $e = b + \xi$ with shared bias b and isotropic noise of scale σ in d dimensions have error second moment with one eigenvalue $\|b\|^2 + \sigma^2$ (bias axis) and $d-1$ eigenvalues σ^2, and participation ratio*

$$\mathrm{PR}(d, \rho) \;=\; \frac{(\rho + d)^2}{(\rho + 1)^2 + (d - 1)}, \qquad \rho = \|b\|^2/\sigma^2,$$

with $\mathrm{PR}(d, 0) = d$, $1 \leq \mathrm{PR} \leq d$, strict decrease in ρ, and the explicit collapse bound $\mathrm{PR} - 1 \leq 3(d-1)/\rho$ for $\rho \geq d$.

The algebra is a one-line corollary of spiked-covariance theory and the participation ratio is a standard effective-dimension metric; the value here is the consilience it enables (§3) and the machine-checked monotonicity that licenses inverting measured PR into a systematic fraction $\alpha = \rho/(\rho + 1)$.

Theorem 4 (Ribbon/consensus decoupling)**.** *The participation ratio of a shared-axis ensemble $\{c_i u\}$ is 1 for every sign pattern of the coefficients c_i, while mean pairwise alignment distinguishes them (e.g. 1 vs. $-1/3$ for three members). PR detects the* axis*; alignment detects* sign coherence*; they are distinct order parameters.*

All four theorems (and the supporting lemmas, ∼29 statements) are machine-checked in Lean 4 against a pinned Mathlib, with zero `sorry` and zero new axioms; the artifact and the theorem-to-statistic contract ship with the replication kit (§10).

3 Layer 1: classical interatomic potentials

The observational base (Welcing, 2026): elastic-constant (C_{11}, C_{12}, C_{44}) errors of 559 classical potentials across 15 cubic metals. Three facts matter for the law. First, error dimensionality is low and *stationary*: participation ratios of multi-element potentials occupy $[1.0, 2.1]$ out of 3 with median 1.28, with no trend across forty years of potential development — accuracy improved; geometry did not. You cannot fit your way off a constraint surface. Second, within a functional-form family the errors are nearly identical (within-family reference–prediction correlation $r = 0.95$ across families vs. 0.82 pooled). Third, the gauge locks: the median PR of 1.28 inverts (Theorem 3, d=3) to a systematic fraction $\alpha \approx 0.93$, independently estimated as 0.95 by within-family correlation and 0.96 by rank-one share — three estimators, one parameter, no tuning. The binding constraint at this layer is the functional form itself; the oldest example is exact: central-force pair potentials satisfy the Cauchy relation $C_{12} = C_{44}$ identically, so for any material violating it, every pair potential's error necessarily contains the same forced component — the constraint dictates the direction.

We emphasize the distinction from parameter-space sloppiness: hyper-ribbon geometry was introduced for the parameter manifolds of *single* models (Transtrum et al., 2011; Machta et al., 2013; Quinn et al., 2023; Kurniawan et al., 2022). The object here is different — error vectors of an ensemble of *distinct* models in observable space — though the two pictures are related through the projection of parameter manifolds into data space.

4 Layer 2: foundation MLIPs — functional vs. architecture

Foundation MLIPs replace the functional-form constraint with expressive neural networks trained on DFT corpora. The law predicts the anisotropy survives and rotates: the binding constraint becomes the training reference theory. Qualitative inheritance of Perdew–Burke–Ernzerhof (PBE) bias — systematic softening shared by universal MLIPs — has been reported (Deng et al., 2025; Yu et al., 2024); the contributions here are the directional quantification and the factorial attribution.

Observation. Three PBE-lineage models of unrelated architecture (MACE-MP-0 (Batatia et al., 2024), CHGNet (Deng et al., 2023), Orb-v3 (Rhodes et al., 2025)) commit *the same* elastic-constant errors on FCC metals against experimental references: all-negative, C_{44}-dominant (Au: $-32\%, -27\%, -79\%$), with cross-architecture cosines 0.95–0.99 for Au, Pt, Ta, Ag. An internal control excludes harness artifacts: Ni, through the identical pipeline, has near-zero error. Across 8–11 models per element the rank-one share of squared error is 0.56–0.94 — and where models "disagree" (V, Cr), they disagree by *sign along the same axis* (cosine -0.88), exactly the structure Theorem 4 distinguishes.

Pre-registered factorial test. The MatPES release (Kaplan et al., 2025) provides one curated training distribution in two functionals (PBE, r^{2}SCAN) with four architectures trained on each (M3GNet, TensorNet, CHGNet, QET) — a 4×2 crossing of constraint against implementation. We registered thresholds and a refutation condition ("errors cluster by architecture") before executing any model, ran all eight cells through one Born-screened strain-energy harness, and obtained: clustering by functional $S_{\text{func}} = +0.317$ (exact permutation $p = 0.029$ over all 70 labelings) versus clustering by architecture $S_{\text{arch}} = -0.093$; the registered rotation prediction confirmed on Au (C_{44} error $-0.33 \to -0.07$) and Pt ($-0.46 \to -0.31$) under r^{2}SCAN; and the dataset-control confirmed (MatPES-PBE models align with MPtrj/OMat-PBE anchors, median cosine 0.66). Two auxiliary thresholds failed as registered (within-functional median 0.654 vs. 0.70; separation 0.085 vs. 0.30): the data revealed *nested* constraints — Cu, Ni, Al errors reorganize by functional (between-functional cosine down to -0.49, the predicted crossover), while the 5d noble metals retain a shared direction across both functionals, bound by correlation physics deeper than the PBE/r^{2}SCAN distinction. The refutation condition was not triggered.

5 Layer 3: DFT implementations — pseudopotential vs. code

One layer further up, the functional itself is held fixed. The ACWF verification effort (Bosoni et al., 2024; Lejaeghere et al., 2016) provides equation-of-state parameters (V_0, B_0, B_1) for 384 unary crystals from eleven pseudopotential-based methods and two all-electron codes, all PBE. Against the all-electron average, with the FLEUR–WIEN2k split as the per-system noise floor, the law predicts errors organize by *pseudopotential table* (the approximation actually shared), not by *simulation code* (the implementation). The dataset's factorial structure makes the test sharp: ABINIT appears with three pseudopotential families, Quantum ESPRESSO and CASTEP with two each, while the PseudoDojo-0.4 table is implemented by four independent plane-wave codes.

Pre-registered outcome: same-table–different-code alignment $S_{\text{table}} = +0.526$ versus same-code–different-family $S_{\text{code}} = +0.265$; separation $+0.261$, permutation $p = 0.017$; refutation condition (clustering by code) not triggered. At pair level, four independent codes sharing PseudoDojo-0.4 align at 0.70–0.87, while ABINIT-with-JTH versus ABINIT-with-PseudoDojo — the same code — drops to 0.21. Two auxiliary predictions failed as registered, both informatively: the within-table consensus median (0.459 vs. 0.50) was dragged down solely by SIESTA, the one localized-orbital-basis code in the group, whose binding constraint is evidently the basis set rather than the shared pseudopotential — the nested-constraint phenomenon again; and the registered regime prediction had the wrong sign because between-family divergence grows on difficult elements, which the pooled statistic conflated. A secondary observation sharpens the law: PAW codes with *different* PAW tables align at only $+0.20$ — the constraint is the specific frozen-core construction, not the formalism label.

Table 1: The projection law across one epistemic stack. Each row: an ensemble of models, the constraint the law identifies as binding, the clustering evidence (constraint vs. implementation), and the registered refutation condition's status.

Layer	Ensemble	Binding constraint	Evidence	Kill con
Classical IPs	559 potentials	functional form	within-family r=0.95; PR invariant 40 yr	(observa
Foundation MLIPs	4×2 MatPES + anchors	training functional	S_{func}=+0.317 vs −0.093, p=0.029	not trig
DFT codes	12 ACWF methods	pseudopotential table	S_{table}=+0.526 vs +0.265, p=0.017	not trig

Scalar reproducibility metrics (Lejaeghere et al., 2016; Bosoni et al., 2024) cannot see any of this structure; it lives in the directions.

6 The conservation–rotation law

Table 1 assembles the stack. Three layers, three different binding constraints, one geometric law; the two new layers tested under pre-registration with explicit kill conditions, neither triggered. Between layers, the direction *rotates*: classical cross-family alignment has no predictive power for MLIP alignment (Spearman $\rho = 0.26$, $p = 0.34$ across elements), because the constraint moved from functional form to training functional; and the MLIP-layer bias direction is precisely the reference-theory error that layer 3 isolates. Within layers, constraints *nest*: when the registered constraint is removed (r^{2}SCAN for PBE; plane-wave pseudopotential sharing for SIESTA), residual alignment reveals the next constraint in line (5d correlation physics; basis set). Anisotropy is conserved throughout: at no layer, in no ensemble, under no intervention did errors approach isotropy — participation ratios remained far below ambient dimension in every case, as Theorem 3 requires of any family whose systematic error dominates its fitting scatter.

The closest antecedents are the persistence of shared biases across climate model generations (Tian and Dong, 2020; Knutti et al., 2013), the convergence of training trajectories onto shared low-dimensional manifolds (Mao et al., 2024), and representation convergence across architectures (Huh et al., 2024); none states conservation-plus-rotation as a law, and none possesses the factorial constraint-vs-implementation evidence presented here.

7 Related work

That ensembles share errors is established: error consistency across networks (Geirhos et al., 2020), model similarity beyond accuracy (Mania et al., 2019), prediction-space clustering (Fort et al., 2019), accuracy- and agreement-on-the-line (Baek et al., 2022). That consensus is not verification is the qualitative lesson of Pirtle et al. (2010) and Parker (2011), operationalized as error correlation by Bishop and Abramowitz (2013); low-dimensional structure of multi-model bias appears in EOF analyses (Sanderson et al., 2015; Brands, 2022). The participation ratio and the spiked-covariance algebra behind Theorem 3 are standard. In materials simulation, shared softening of universal MLIPs is documented (Deng et al., 2025; Yu et al., 2024), reproducibility of DFT codes is measured by scalar metrics (Lejaeghere et al., 2016; Bosoni et al., 2024), and parameter-space sloppiness of single potentials is well developed (Transtrum et al., 2011; Kurniawan et al., 2022; Quinn et al., 2023). Relative to all of the above, the contributions here are: the error-vector geometry of model ensembles in observable space as the measured object; the factorial constraint-vs-implementation designs and their pre-registered outcomes; the conservation–rotation law; and the machine-checked theory chain with a two-tier replication kit.

8 Implications

Calibration without retraining. Because the shared error is one direction, one correction vector per (element, observable) repairs every model in the family at once: a rank-one correction removes ∼96% (median) of foundation-MLIP elastic error. The correction is family-level intellectual property — it transfers to models not yet trained, so long as they share the constraint.

A trust rule for ensembles. Cross-model alignment is a zero-cost diagnostic: high alignment ⇒ the ensemble is constraint-bound, calibratable, and its spread *understates* uncertainty; low alignment at the noise floor ⇒ genuine convergence (the Ni control); low alignment far above the floor ⇒ the constraint is heterogeneous (magnetic BCC metals; difficult elements across pseudopotential tables) and no shared correction exists.

Benchmark design. Leaderboards that pool across families average away the very direction that identifies what to fix. Reporting error *directions* stratified by candidate constraints — the factorial pattern used here — converts benchmarking from scoring into diagnosis.

Reading consensus. Where ground truth is delayed (climate projections, materials discovery), inter-model agreement should be priced as a measurement of shared constraint; the factorial designs show how to estimate *which* constraint, today, from models alone.

9 Limitations

The formal chain covers convex reachable sets and deterministic bias-plus-noise spectra; smooth non-convex families (local normal cones) and the finite-sample step from noisy ensembles to the exact second moment are standard but unformalized. The MLIP layer uses one validated harness, fifteen cubic elemental metals, and elastic observables; the DFT layer inherits ACWF's protocol choices. Four of nine registered auxiliary thresholds across the two experiments failed and are reported as failures; in each case post-hoc analysis attributes the miss to nested constraints, and those attributions are hypotheses for the next registration round, not confirmed claims. The law itself makes its next predictions cheap to test: an r^{2}SCAN-trained model from a fourth architecture must join the r^{2}SCAN cluster; a plane-wave implementation of any new pseudopotential table must align with its table, not its code; and axis-based statistics (mandated by Theorem 4) should replace signed-cosine thresholds.

10 Reproducibility

Both experiments were pre-registered with thresholds and refutation conditions committed to version control before execution (commits `dffbe595` and `ebf39e33`); all analysis is replayable from a two-tier kit in which Tier 1 verifies every statistic in this paper from committed raw data using only NumPy/SciPy in seconds, and Tier 2 re-derives the raw elastic constants from public checkpoints through a frozen harness (bit-exact on revalidation). The Lean 4 artifact (four theory files, ∼29 machine-checked statements, zero `sorry`, zero new axioms, pinned Mathlib) and the theorem-to-statistic contract accompany the kit. The harness was validated by independent energy- vs. stress-method agreement ($\leq 2.4\%$), bit-exact regression, and strain-window stability. ACWF data are from the public archive of Bosoni et al. (2024); MatPES checkpoints from Kaplan et al. (2025).

Acknowledgments

This work was supported by Lupine Science. AI assistance (Anthropic Claude) was used under the author's direction for statistical verification and recomputation, Lean proof engineering, literature triage, and drafting; all scientific claims and decisions were reviewed and approved by the author, who takes sole responsibility for the content.

References

James D. Annan and Julia C. Hargreaves. Understanding the CMIP3 multimodel ensemble. *Journal of Climate*, 24:4529–4538, 2011.

Christina Baek, Yiding Jiang, Aditi Raghunathan, and J. Zico Kolter. Agreement-on-the-line: Predicting the performance of neural networks under distribution shift. In *Advances in Neural Information Processing Systems 35*, 2022.

Ilyes Batatia et al. A foundation model for atomistic materials chemistry, 2024.

Craig H. Bishop and Gab Abramowitz. Climate model dependence and the replicate Earth paradigm. *Climate Dynamics*, 41:885–900, 2013.

Emanuele Bosoni et al. How to verify the precision of density-functional-theory implementations via reproducible and universal workflows. *Nature Reviews Physics*, 6:45–58, 2024.

Swen Brands. Common error patterns in the regional atmospheric circulation simulated by the CMIP multi-model ensemble. *Geophysical Research Letters*, 49:e2022GL101446, 2022.

Bowen Deng, Peichen Zhong, KyuJung Jun, Janosh Riebesell, Kevin Han, Christopher J. Bartel, and Gerbrand Ceder. CHGNet as a pretrained universal neural network potential for charge-informed atomistic modelling. *Nature Machine Intelligence*, 5:1031–1041, 2023.

Bowen Deng, Yunyeong Choi, Peichen Zhong, Janosh Riebesell, Shashwat Anand, Zhuohan Li, KyuJung Jun, Kristin A. Persson, and Gerbrand Ceder. Systematic softening in universal machine learning interatomic potentials. *npj Computational Materials*, 11:9, 2025.

Frank Deutsch. *Best Approximation in Inner Product Spaces*. Springer, New York, 2001.

Stanislav Fort, Huiyi Hu, and Balaji Lakshminarayanan. Deep ensembles: A loss landscape perspective, 2019.

Robert Geirhos, Kristof Meding, and Felix A. Wichmann. Beyond accuracy: quantifying trial-by-trial behaviour of CNNs and humans by measuring error consistency. In *Advances in Neural Information Processing Systems 33*, 2020.

Minyoung Huh, Brian Cheung, Tongzhou Wang, and Phillip Isola. The platonic representation hypothesis. In *Proceedings of the 41st International Conference on Machine Learning*, 2024.

Yiding Jiang, Vaishnavh Nagarajan, Christina Baek, and J. Zico Kolter. Assessing generalization of SGD via disagreement. In *International Conference on Learning Representations*, 2022.

Aaron D. Kaplan et al. A foundational potential energy surface dataset for materials, 2025.

David Kinderlehrer and Guido Stampacchia. *An Introduction to Variational Inequalities and Their Applications*. Academic Press, New York, 1980.

Reto Knutti, David Masson, and Andrew Gettelman. Climate model genealogy: Generation CMIP5 and how we got there. *Geophysical Research Letters*, 40:1194–1199, 2013.

Yonatan Kurniawan, Cody L. Petrie, Kinamo J. Williams, Mark K. Transtrum, Ellad B. Tadmor, Ryan S. Elliott, Daniel S. Karls, and Mingjian Wen. Bayesian, frequentist, and information geometric approaches to parametric uncertainty quantification of classical empirical interatomic potentials. *Journal of Chemical Physics*, 156:214103, 2022.

Kurt Lejaeghere et al. Reproducibility in density functional theory calculations of solids. *Science*, 352:aad3000, 2016.

Benjamin B. Machta, Ricky Chachra, Mark K. Transtrum, and James P. Sethna. Parameter space compression underlies emergent theories and predictive models. *Science*, 342:604–607, 2013.

Horia Mania, John Miller, Ludwig Schmidt, Moritz Hardt, and Benjamin Recht. Model similarity mitigates test set overuse. In *Advances in Neural Information Processing Systems 32*, 2019.

Jialin Mao, Itay Griniasty, Han Kheng Teoh, Rahul Ramesh, Rubing Yang, Mark K. Transtrum, James P. Sethna, and Pratik Chaudhari. The training process of many deep networks explores the same low-dimensional manifold. *Proceedings of the National Academy of Sciences*, 121:e2310002121, 2024.

Wendy S. Parker. When climate models agree: The significance of robust model predictions. *Philosophy of Science*, 78:579–600, 2011.

Zachary Pirtle, Ryan Meyer, and Andrew Hamilton. What does it mean when climate models agree? A case for assessing independence among general circulation models. *Environmental Science & Policy*, 13:351–361, 2010.

Katherine N. Quinn, Michael C. Abbott, Mark K. Transtrum, Benjamin B. Machta, and James P. Sethna. Information geometry for multiparameter models: new perspectives on the origin of simplicity. *Reports on Progress in Physics*, 86:035901, 2023.

Benjamin Rhodes, Sander Vandenhaute, Vaidotas Šimkus, James Gin, Jonathan Godwin, Tim Duignan, and Mark Neumann. Orb-v3: atomistic simulation at scale, 2025.

Benjamin M. Sanderson, Reto Knutti, and Peter Caldwell. A representative democracy to reduce interdependency in a multimodel ensemble. *Journal of Climate*, 28:5171–5194, 2015.

Baijun Tian and Xinyu Dong. The double-ITCZ bias in CMIP3, CMIP5, and CMIP6 models based on annual mean precipitation. *Geophysical Research Letters*, 47:e2020GL087232, 2020.

Mark K. Transtrum, Benjamin B. Machta, and James P. Sethna. Geometry of nonlinear least squares with applications to sloppy models and optimization. *Physical Review E*, 83:036701, 2011.

Alex Welcing. The causal geometry of prediction errors in interatomic potentials: a hyper-ribbon manifold analysis with ecological fallacy detection. Submitted to Integrating Materials and Manufacturing Innovation, 2026.

Haochen Yu, Matteo Giantomassi, Giuliana Materzanini, Junjie Wang, and Gian-Marco Rignanese. Systematic assessment of various universal machine-learning interatomic potentials. *Materials Genome Engineering Advances*, 2:e58, 2024.